

CSE471

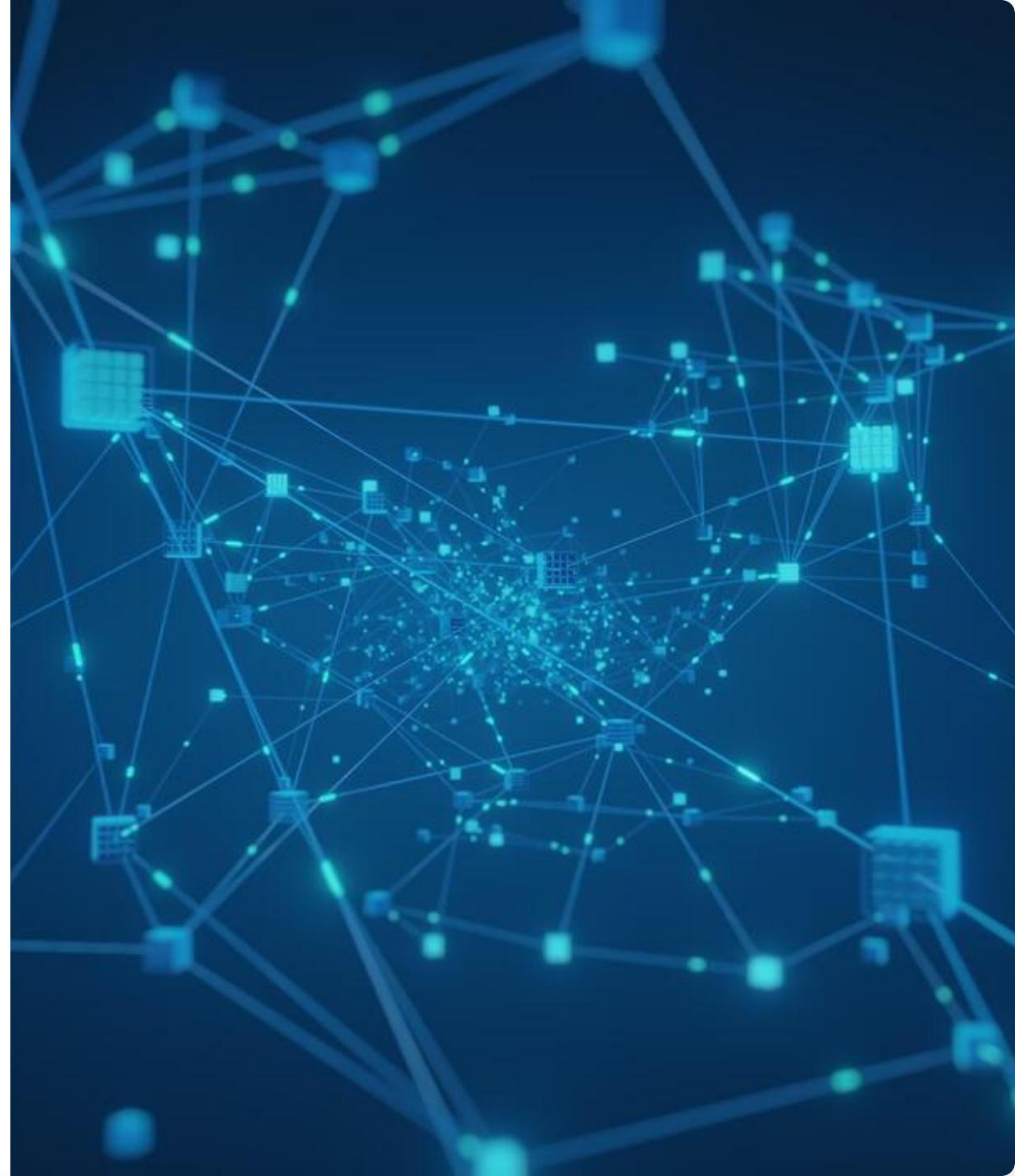
Deep Learning for Computer Vision

Lecture 1

Why Computer Vision Matters: Traditional vs Deep Learning

Lovely Professional University

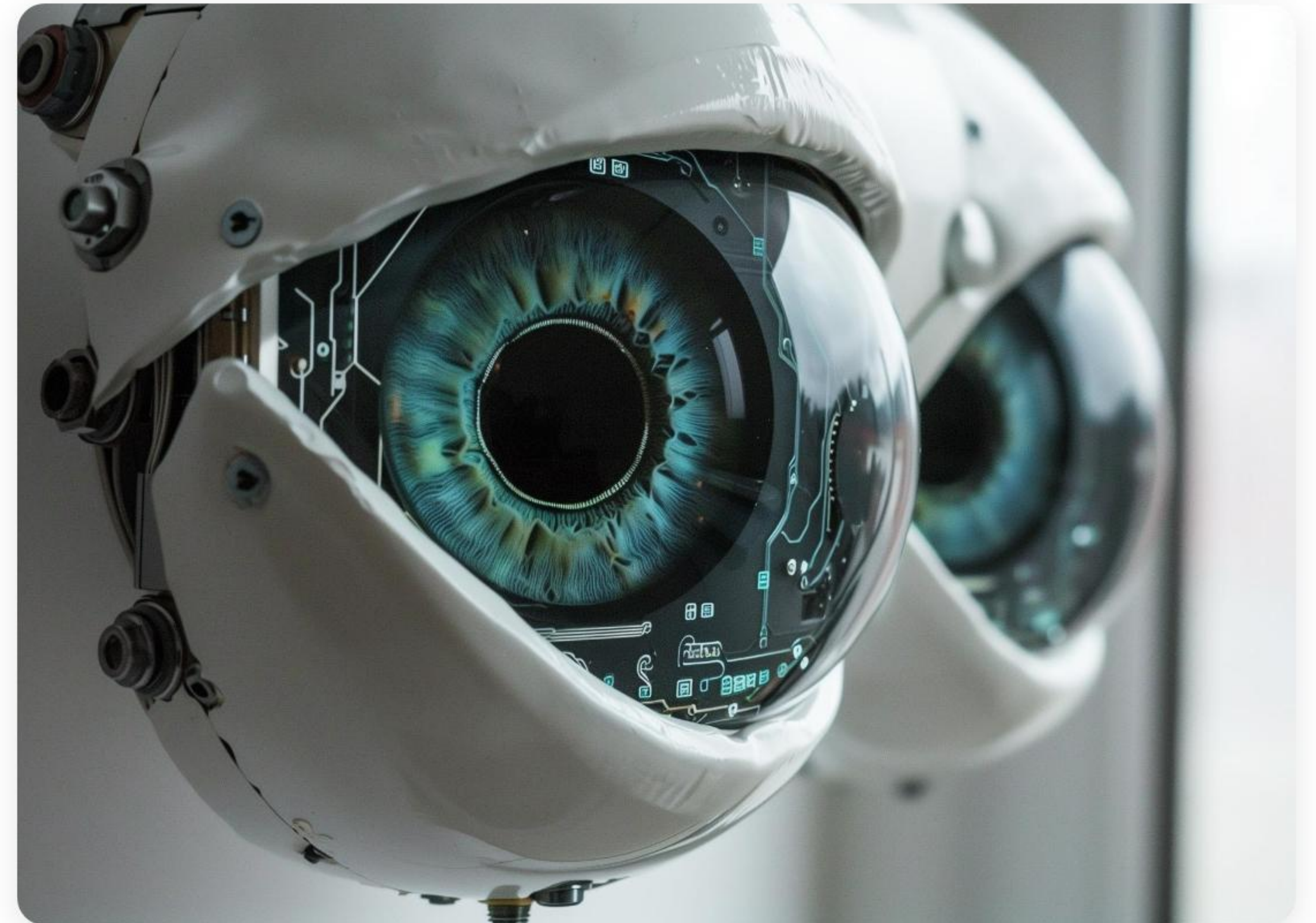
Instructor: Prof. Rishi Raj



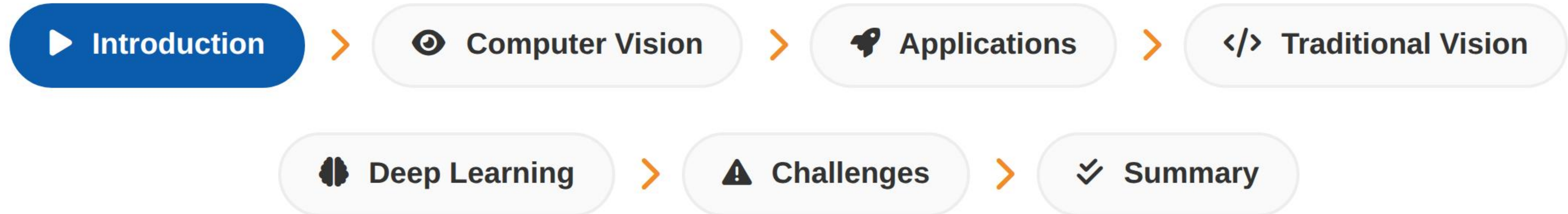
Learning Outcomes

By the end of this lecture, you should be able to:

- ▶ Explain the core concept of Computer Vision.
- ▶ Understand why machines need visual perception.
- ▶ Identify major real-world applications.
- ▶ Compare traditional and deep learning methodologies.
- ▶ Differentiate Image Processing from Computer Vision.



Lecture Roadmap



SET INDUCTION

The Driving Analogy

"How does a computer understand this scene?"

Identify critical elements:

🚗 Drivable Road

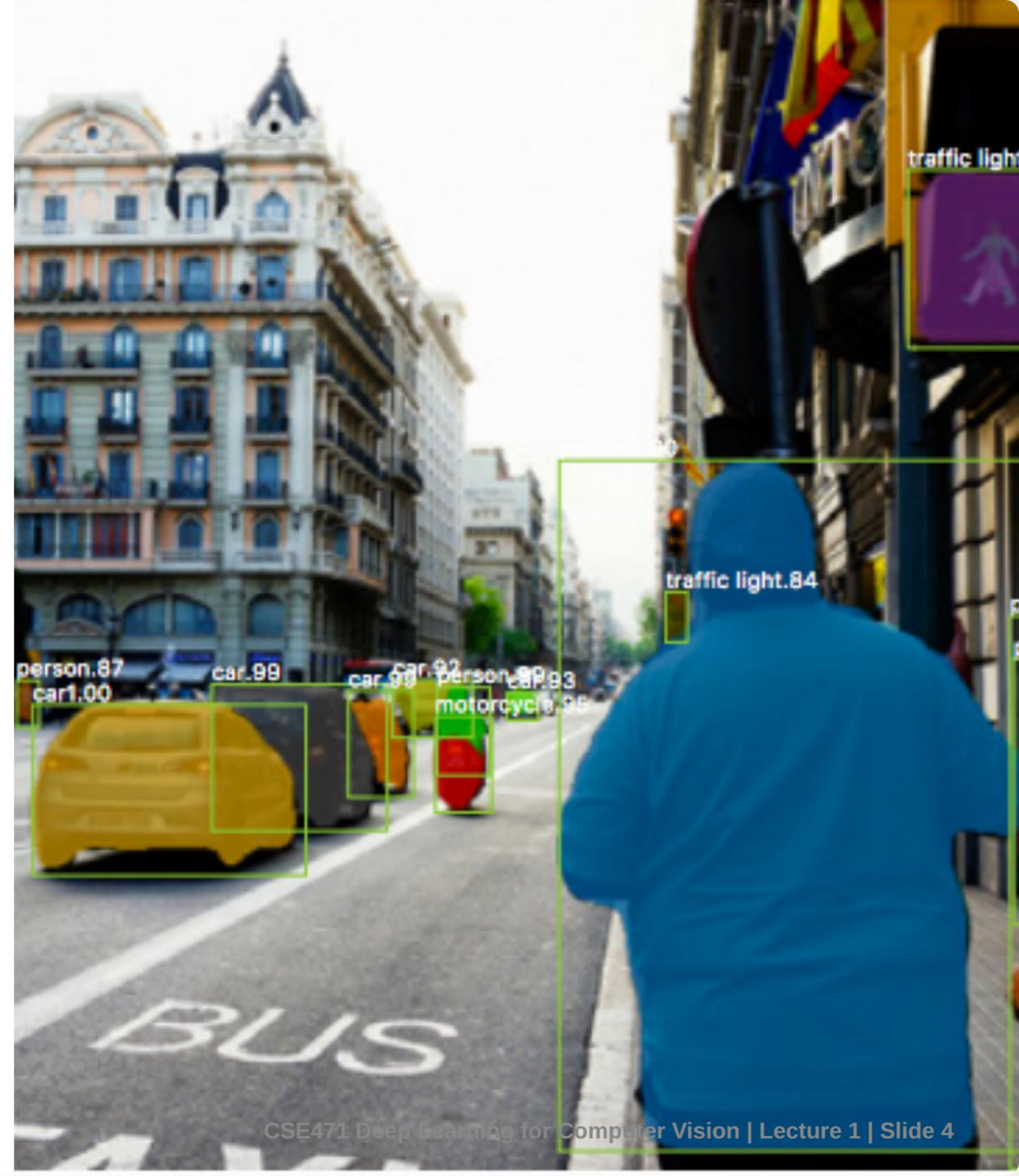
🚗 Vehicles

🚲 Cyclists

🚦 Traffic Lights

🚶 Pedestrians

🚦 Road Signs



What is Computer Vision?

Computer Vision is a branch of Artificial Intelligence that enables computers to understand and interpret visual information from images and videos.



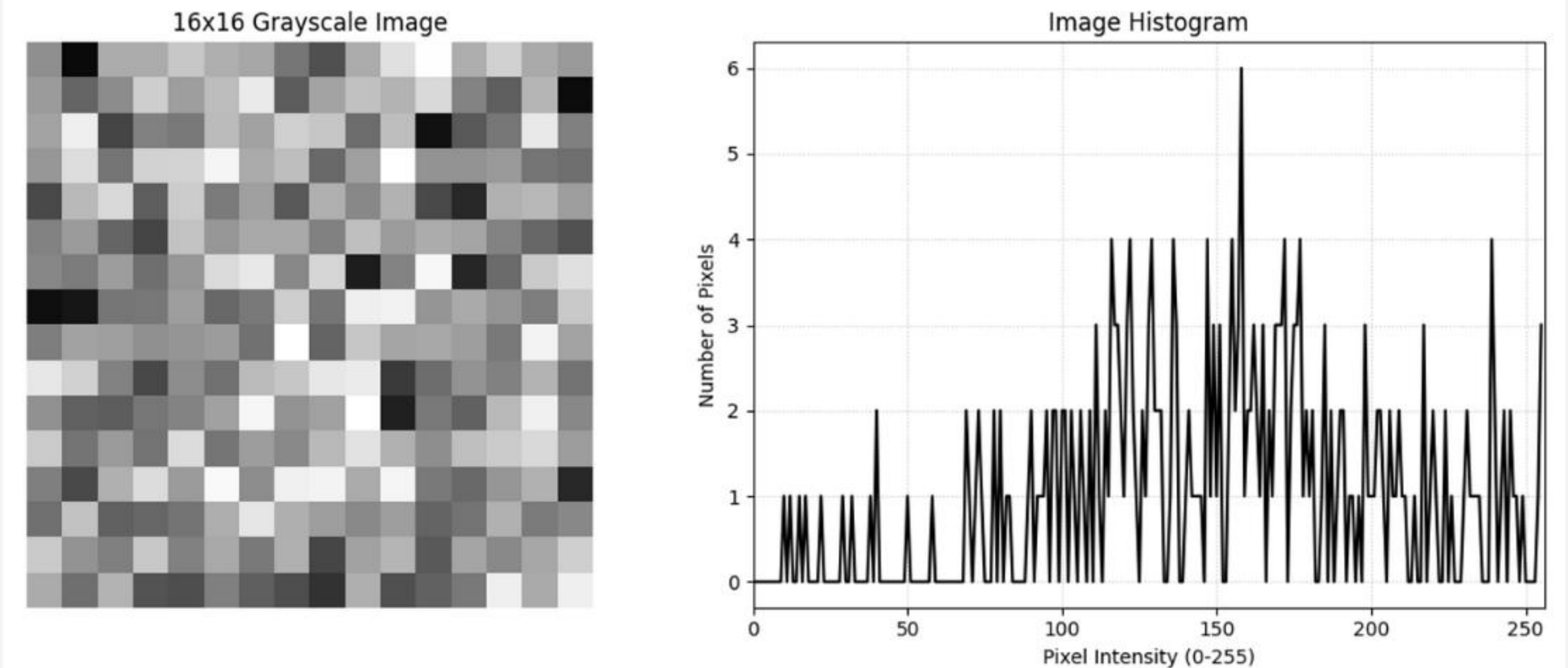
Why is Computer Vision Difficult?

The Semantic Gap

Humans instantly recognize objects, context, and depth based on years of natural learning.

Computers initially perceive only a massive 2D or 3D grid of numerical pixel values (intensities).

$f(x, y) = [R, G, B]$
[255, 128, 0] $\rightarrow$???



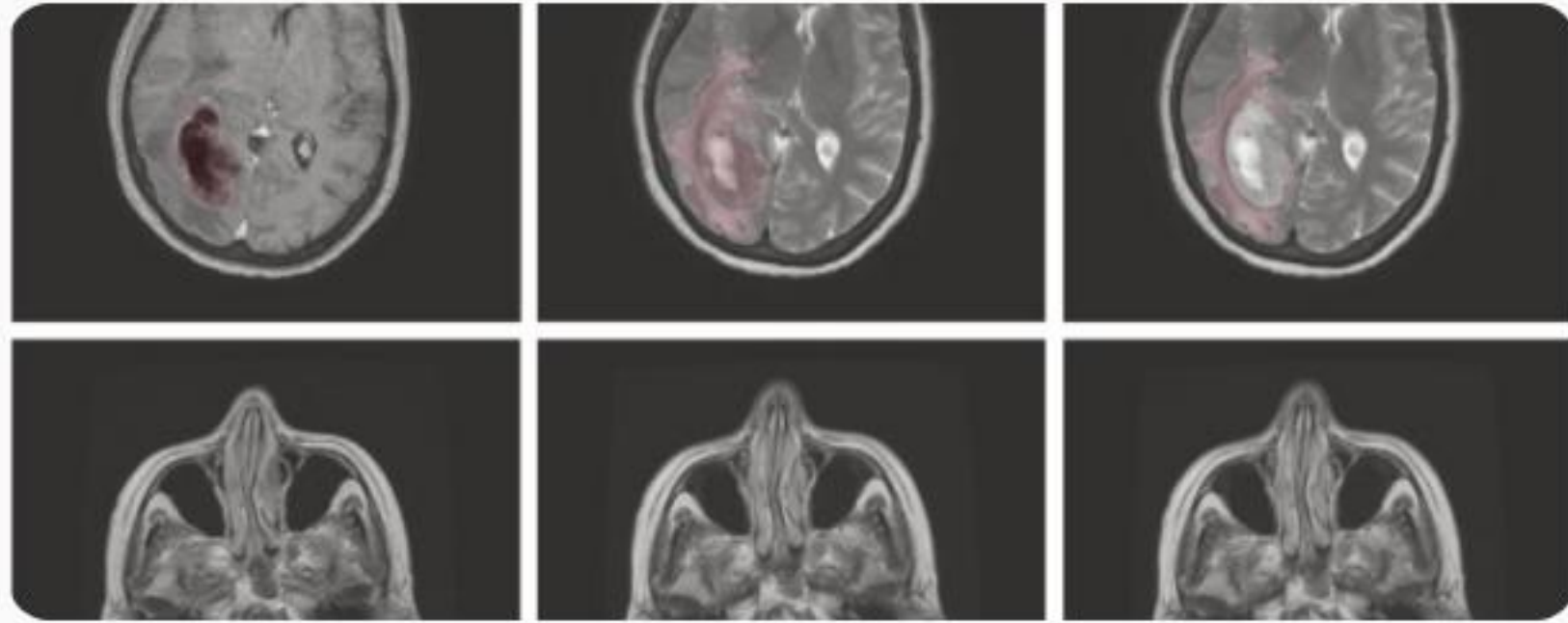
Human Vision vs Computer Vision

☰ Feature	👁 Human Vision	📷 Computer Vision
Sensor	Biological Eyes (Retina)	Digital Camera (CMOS / CCD)
Processor	Visual Cortex (Brain)	CPU / GPU / TPU / Algorithms
Learning	Natural, continuous, unsupervised	Dataset-driven, mostly supervised
Generalization	Excellent (handles novel scenes easily)	Struggles with domain shifts
Adaptability	Highly Robust to lighting/occlusion	Brittle without augmented data

Why Computer Vision Matters



Real-world Applications



Healthcare

Tumor Detection, MRI, X-rays Analysis



Transportation

Autonomous Vehicles, Traffic Recognition



Agriculture

Crop Disease Detection, Yield Estimation



Retail

Cashier-less Stores, Inventory Tracking



Manufacturing

Automated Defect Detection (QA/QC)



Social Media

Face Filters, Auto Tagging, Deepfakes

Classroom Activity: Vision in Your Pocket

How many apps on your smartphone use Computer Vision?



Instagram



Google Lens



Snapchat



Face Unlock



WhatsApp

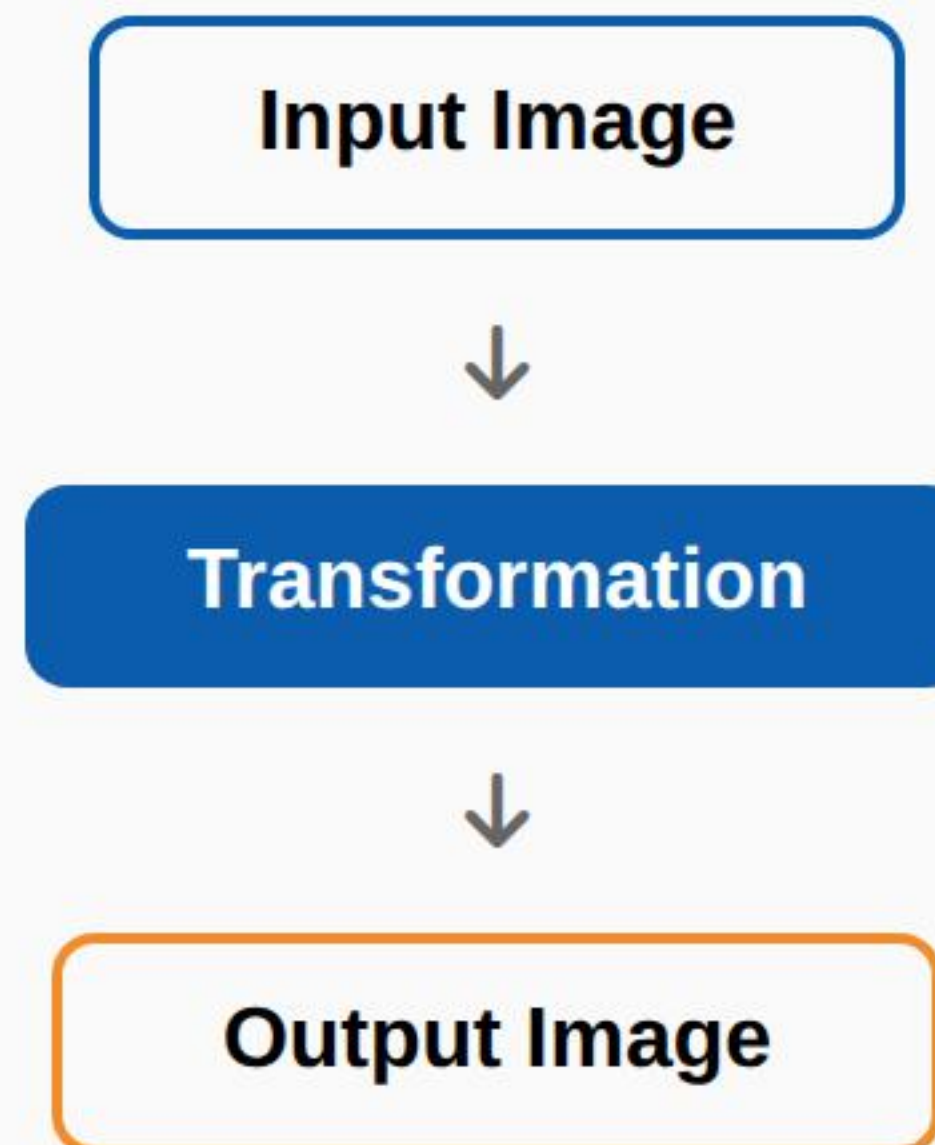


QR Scanner

Discuss with the person next to you for 2 minutes.

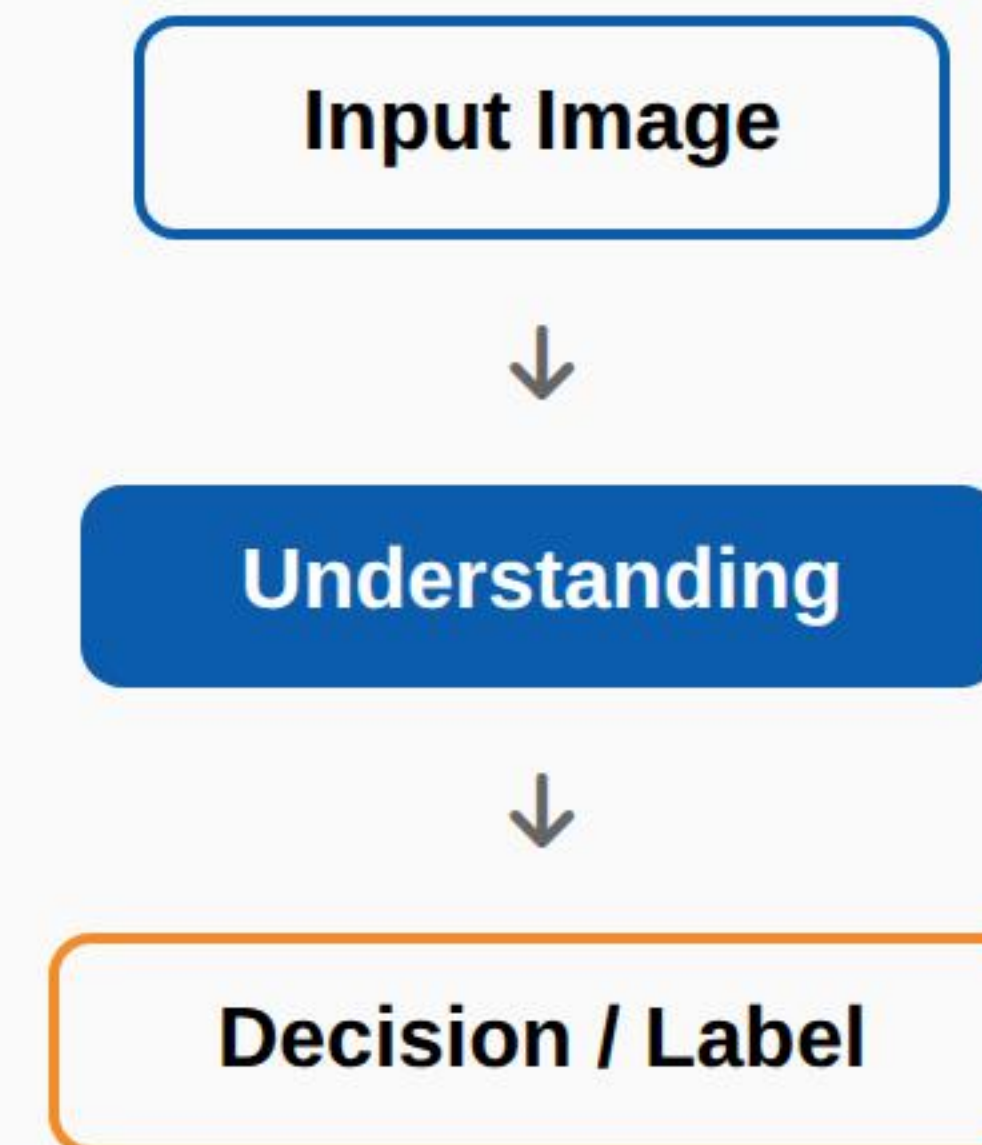
Image Processing vs Computer Vision

Image Processing



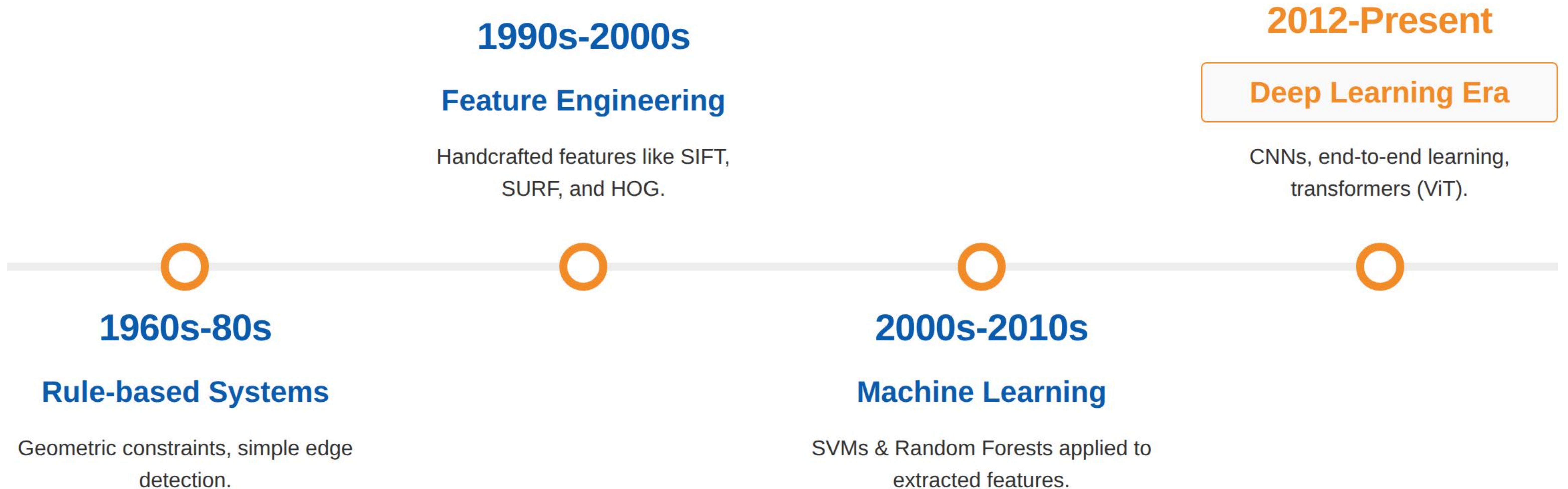
- ▶ Blur / Sharpen
- ▶ Noise Removal
- ▶ Contrast Enhancement

Computer Vision



- ▶ Face Recognized
- ▶ Tumor Classified
- ▶ Vehicle Detected

Evolution of Computer Vision



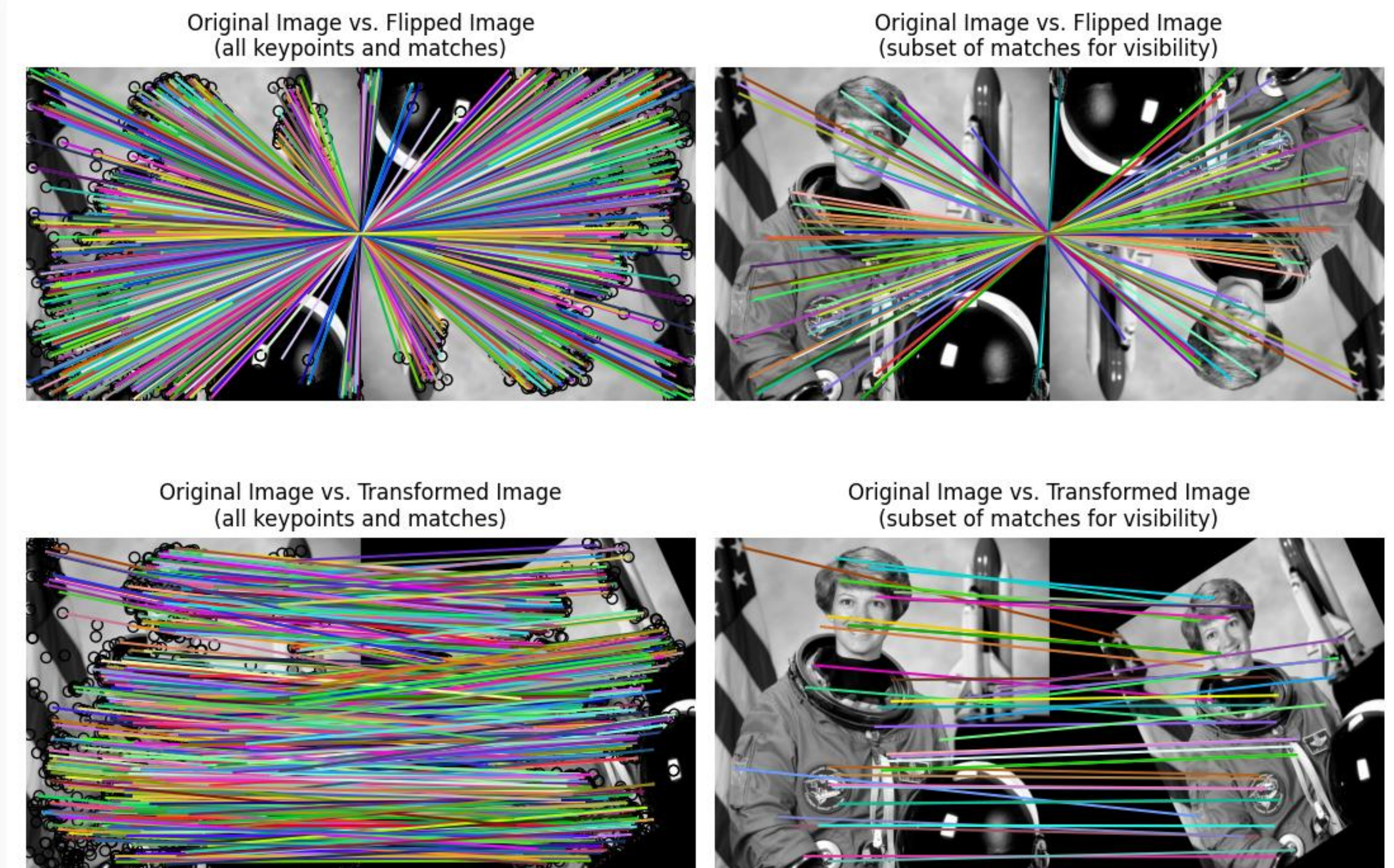
The Feature Engineering Era

Before Deep Learning, experts mathematically defined algorithms to find **interest points**.

- ▶ **SIFT:** Scale-Invariant Feature Transform
- ▶ **SURF:** Speeded Up Robust Features
- ▶ **HOG:** Histogram of Oriented Gradients
- ▶ **LBP:** Local Binary Patterns

⚠ Limitations:

Highly manual, mathematically complex, and brittle to large variations in lighting or occlusion.



Traditional Machine Learning Pipeline



Common Classifiers Used:

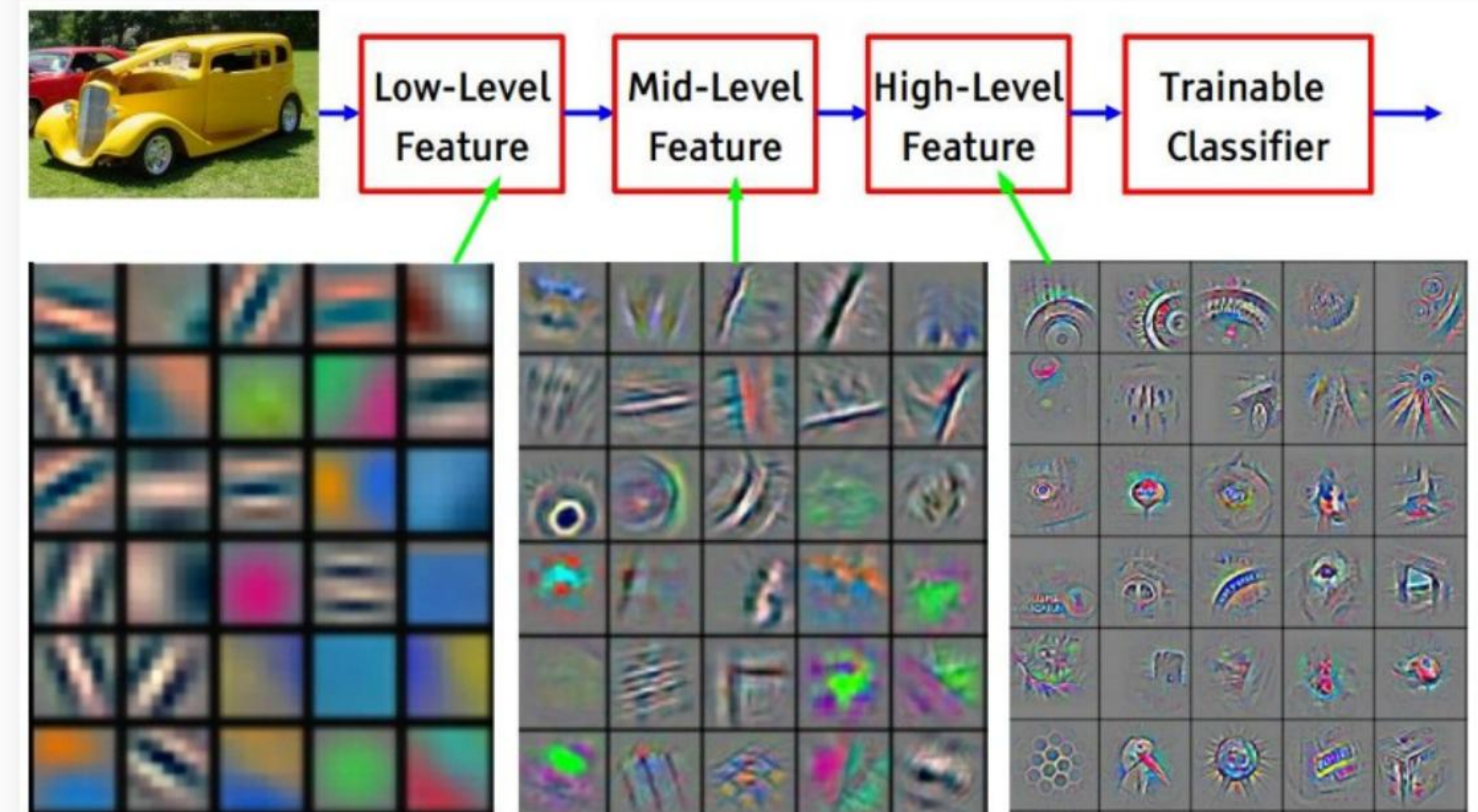
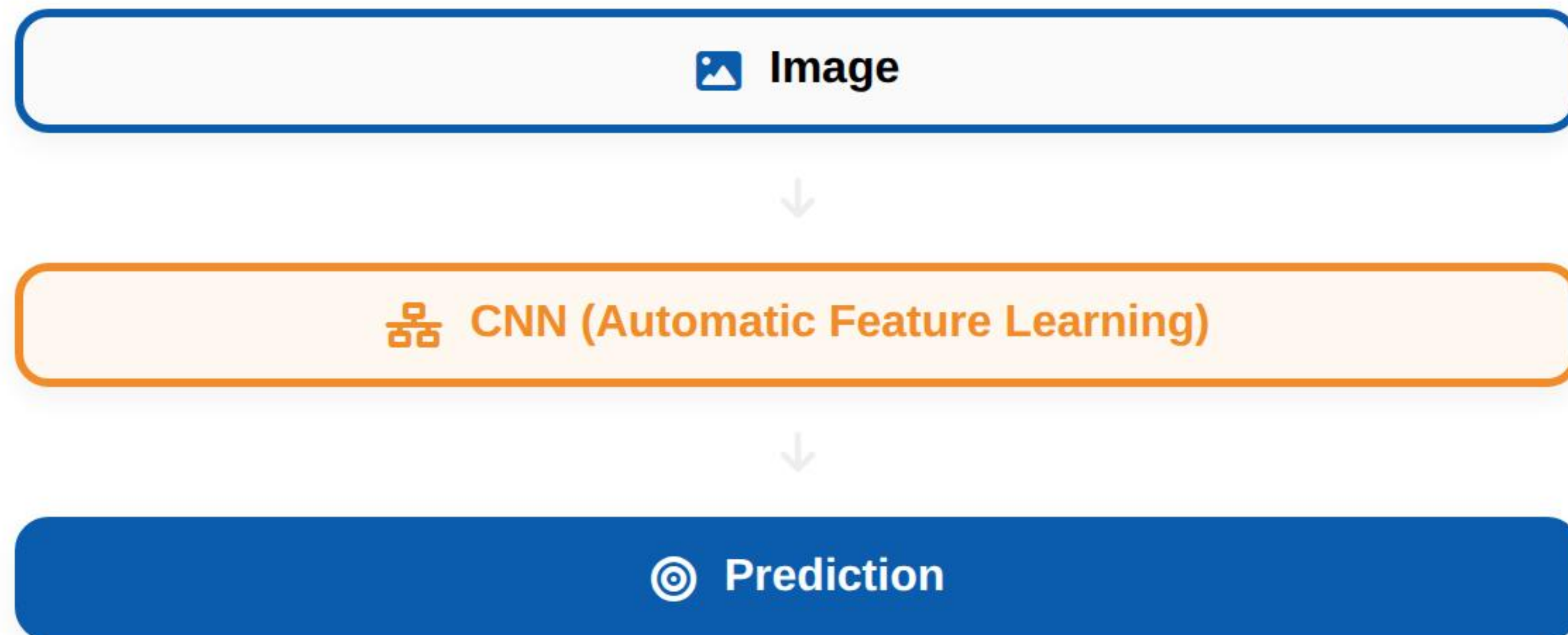
Support Vector Machines (SVM)

Random Forest

Decision Trees

Deep Learning Pipeline

End-to-End Learning: We no longer write mathematical rules to find edges or corners.



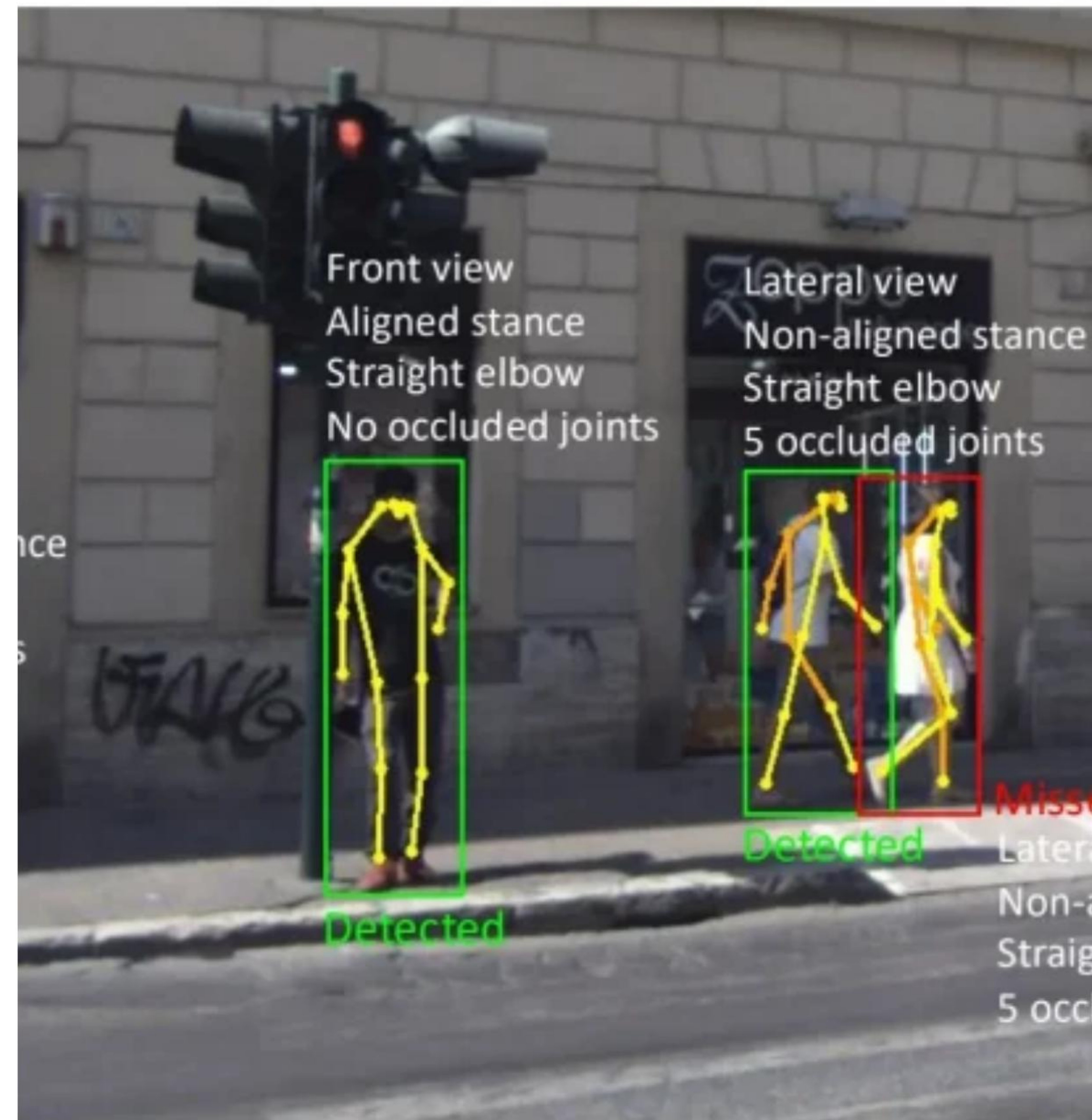
Traditional CV vs Deep Learning

Metric	Traditional ML	Deep Learning
Feature Extraction	Manual (Handcrafted)	Automatic (Learned)
Data Dependency	Works well on small datasets	Requires massive datasets
Architecture	Multiple disconnected stages	End-to-End optimization
Hardware Compute	CPU is usually sufficient	GPU / TPU highly intensive
Accuracy / Scalability	Plateaus quickly	Scales incredibly well

Challenges in Computer Vision

Despite deep learning breakthroughs, CV models still struggle with natural variations:

- 💡 Lighting Variations
- 👁️ Occlusion & Blur
- 📹 Different Camera Angles
- 📶 Background Clutter
- 🚶 Intra-class Pose Variation
- 🔍 Scale Variation



Summary & Key Takeaways

- ✓ Computer Vision helps machines map pixels to semantic meaning.
- ✓ Traditional methods depend heavily on brittle, handcrafted features.
- ✓ Deep Learning revolutionized the field by learning features automatically.
- ✓ CV is the foundational technology powering modern AI systems globally.

Think – Pair – Share

Question:

Can safe and reliable autonomous vehicles be built using *only* traditional Computer Vision techniques?

🕒 Discuss with your partner for 3 minutes. Be prepared to share your reasoning.

Homework & References

Homework

Find 5 real-world applications of Computer Vision. For each, classify the underlying task:

- ▶ Image Classification
- ▶ Object Detection
- ▶ Semantic Segmentation
- ▶ Instance Segmentation
- ▶ Pose Estimation
- ▶ OCR / Face Recognition

Due: Next Lecture via Canvas.

References

- ▶ Szeliski, R. *Computer Vision: Algorithms and Applications*.
- ▶ Goodfellow, I. et al. *Deep Learning*.
- ▶ Stanford University: CS231n Course Materials.
- ▶ PyTorch & TensorFlow Official Documentation.
- ▶ OpenCV Official Documentation.

Image Sources



https://upload.wikimedia.org/wikipedia/en/3/3a/Lovely_Professional_University_logo.png

Source: en.wikipedia.org



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Source: www.vecteezy.com



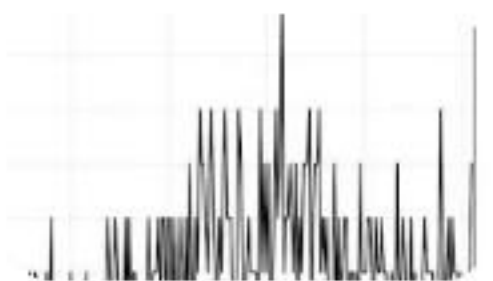
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Source: scitechdaily.com



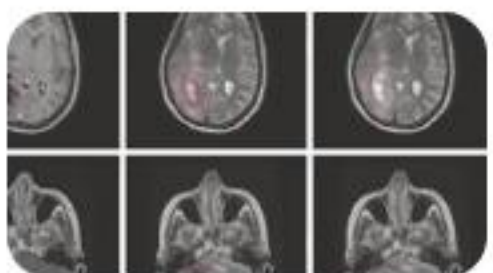
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Image Sources



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Source: stackoverflow.com



<https://objectways.com/wp-content/uploads/2025/11/pose-estimation-challenges-with-occluded-joints-1024x478.webp>

Source: objectways.com